1 A Hyper-Local Electricity Demand Forecasting Framework Using LSTM, Prophet, and Ensemble Models

Abstract

This paper will describe a study aimed at predicting short-term demand for electricity on a very small scale to assist in operating and planning for the electric grid. This research will include the analysis of electric loads over historical periods and combining this with the weather forecast to create a short-term (48-hour) electric forecast. The information used for this research will be publicly available electric system data from several countries and their corresponding weather data. Using a method of advanced time series forecasting known as Long Short-Term Memory (LSTM) (neural network), the model captures the time-based patterns of electric load forecasting. By integrating additional engineering techniques with LSTM, such as lagging electric demand, statistical uses of electric demand, extinct device forecasting, as well as weather variables including temperature, humidity, wind speed, and atmospheric pressure, it is anticipated that the LSTM forecasts will outperform the results of a baseline forecast using the Prophet algorithm. The use of LSTM as the primary forecast will provide significant improvements in forecasting accuracy for the short-term loading of electric grid systems and, therefore, presents a viable and scalable method to predict short-term electric loads for emerging smart grid systems.

Keywords: Electricity demand forecasting, short-term load forecasting, hyper-local forecasting, time-series analysis, deep learning, Long Short-Term Memory (LSTM), weather-based forecasting, smart grid, feature engineering, energy analytics.

Introduction

Forecasting electricity demand is an important part of the operation of a reliable electricity supply and managing how the utility system allocates electrical resources efficiently and at the lowest possible cost. The rapid growth of urban life, along with the growing number of buildings using electricity and the integration of renewable sources into the electricity supply, has made controlling and operating today's electricity supply systems much more complex. Thus, accurate, short-term electricity demand forecasts will contribute to fully implementing the smart grid (SG), demand response (DR), and real-time operational decisions.

Patterns of electricity demand are often described as "temporal" and "historical" in nature and are established by external, calendar-based (time of day, day of week, seasonality), and consumer-driven (i.e., behavioral) parameters. As well as these calendar-based parameters, weather conditions and other external variables also significantly influence short-term electricity demand. Measurements of temperature, and degrees of humidity, wind speed, and pressure in the atmosphere can impact heating and cooling loads and subsequently lead to higher amounts of energy consumption. Therefore, it is essential to include weather-related variables in electricity demand forecasts to increase their accuracy level.

Due to the interpretability and simplicity of traditional statistical and regression-based methods of short-term load forecasting, these types of analyses are a common choice for short-term load forecasting. Unfortunately, many of these approaches struggle to adequately model the nonlinear and dynamic relationships between the determinants of electricity demand and electricity usage, particularly when using high-frequency (hourly) data. On the other hand, many machine learning techniques have improved forecasting results by modeling nonlinear patterns; however, the majority of machine learning techniques are restricted in their capacity to learn long-term temporal dependencies from time-series data.

Recently, Long Short-Term Memory (LSTM) networks, which are a type of deep learning algorithm, have shown promising results for short-term forecasting of electricity demand. LSTMs are built to specifically capture long-term temporal dependencies and have outperformed both traditional statistical and machine-learning-based approaches to short-term forecast electricity demand in the energy domain. Another alternative for forecasting short-term electricity demand includes using time-series models, such as Prophet, as a straightforward and interpretable benchmark to capture both trend and seasonality related to predicting short-term electricity demand.

Most of the current research on forecasting has been centered around either large-scale or state/national level models and does not provide much in the way of information regarding hyper-local demand forecasting based on fine-scale data from the weather. In this paper, we will try to fill some of that gap by developing a framework for forecasting electricity demand at the hyperlocal level using an LSTM deep-learning model incorporated into an approach that utilizes weather-based features for developing short-term (up to the next 48 hrs) hourly forecasts of demand. Additionally, we will develop a baseline model using Prophet so that we can compare the results and performance of the proposed approach. We will evaluate the performance of the proposed methodology using publicly available electricity load data and weather datasets, and will show that the proposed methodology is suitable for providing hyper-localized short-term forecasts of electricity demand using modern versions of the smart grid.

Literature Review

Hybrid deep learning models for short-term electricity load forecasting

Hybrid modeling frameworks that combine statistical decomposition of time-series data with machine learning methods, like deep learning, have been increasingly used in short-term forecasting of residential electricity usage. As a case in point, Bao and Lu (2024) found that when they incorporated NeuralProphet into a neural network architecture consisting of CNN and LSTM, they achieved forecasting accuracy that was superior to forecasting either as standalone components, yet they offered minimal exploration of how fine-grained weather observations can enhance hyper-local forecasting. Also, Shohan et al. (2022) combined NeuralProphet with an LSTM architecture to better capture the trends, seasonality, and nonlinear dependence of electricity load over different periods of time, and reported that they had made superior short-term forecast accuracy; however, much of their model depended exclusively on the historical usage of electricity, and did not adequately utilize high-resolution weather datasets. In summary, despite the advantages of hybrid models to improve forecasting accuracy as illustrated in previous research studies, the use of hybrid models with respect to forecasting hyper-local electricity load using hyper-local weather conditions is limited.

H1: *Hybrid NeuralProphet–deep learning models do not significantly improve hyper-local electricity load forecasting accuracy.*

Comparative performance of LSTM and Prophet models in electricity demand forecasting

Some studies have compared the performance of LSTMs to Prophet-based methods for electricity demand forecasting. Albahli (2025) compared the performance of LSTMs and Prophet models in terms of forecast accuracy and performance across varying demand conditions and found that while Prophet models have good performance in stable demand scenarios (with patterns reasonably stable), LSTM models tend to have superior performance in more complex, non-linear demand conditions. Albahli's study included a detailed analysis of the strengths and weaknesses of both model types, but did not evaluate their performance in relation to hyper-local forecasting needs, weather-driven feature engineering, or deployment-oriented forecasting pipelines for localized energy management. The empirical evidence available supports the conclusion that, while Prophet can capture the trend and seasonal component of demand, LSTMs are better suited to model the nonlinear temporal dynamics of demand than Prophet. The literature also supports a complementary relationship between LSTM and Prophet methods, which motivates the need to integrate these two methods to enhance the accuracy of short-term forecasts at finer spatial resolutions.

H2: *LSTM-based models outperform Prophet models in short-term electricity demand forecasting under dynamic and nonlinear conditions.*

Hybrid Prophet-GRU models for household electricity consumption forecasting

According to Prabavathi and Kanmani (2024), they developed a mixed-prophet-gru analysis method that combines both Prophet (which detects and represents seasonal and long-term changes) with a GRU model that detects time-based dependencies in data to predict how much electricity is used in a home. The researchers found that this combined-prophet/gru hybrid provided much more accurate short-term predictions of how much electricity will be used compared to using either Prophet or GRU alone. Nonetheless, as their analysis was conducted using data from only one household, they did not compare this hybrid approach to other deep learning techniques available today, such as LSTM for making predictions based on multiple locations, nor have they incorporated detailed weather-related variables into their model when making these predictions.

H3: *Hybrid Prophet–GRU models do not significantly improve hyper-local household electricity demand forecasting accuracy.*

Data and Variables

Study Period and Sample

For five years (January 2018 - December 2022), the study will take place, with hourly frequency and short-term electricity demand forecasting. It will examine the newest electricity consumption trends, seasonal effects, and how climatic and socio-economic variables have changed electricity consumption. Approximately 43,800 hourly total electricity load observations will be used, providing enough data to model both short- and long-term variations and trends. The study will utilize only publicly available electricity load data, which were collected in a time series format with the appropriate granularity and sufficiently large sample size to be representative of the population. In addition to the above, the researchers also used publicly available weather data (temperature, humidity, wind speed, and atmospheric pressure) to perform the analysis. All datasets used will be chronologically partitioned into training, validation, and test datasets to obtain realistic conditions for forecasting. The last 48 hours of the entire dataset were reserved for a test dataset, the 7 days before that for the validation dataset, and the remaining observations will be used to train the model. The sampling and evaluation methodology ensured that the obtained forecasting performance for short-term forecasting was valid.

Dependent Variable

Electricity demand is the total consumption of electricity in a given area over a number of hours, and will be the main target variable representing the association between how much electricity customers use and how their usage changes due to things like the time of year or the weather.

The independent variables used for this study are historical electricity demand and meteorological (i.e., weather) variables, such as temperature, humidity, wind speed, and atmospheric pressure. The lagged (previous) values of electricity load are also being used so that the model captures the temporal relationship that exists within the data. Moreover, the use of rolling statistical features should help to provide a better prediction than if they were excluded.

In addition, there are control variables being used in this model. Calendar-based features, such as hour of day, day of week, and public holidays, will all be included as control variables to estimate and eliminate the impact of periodic and structural variations on electricity consumption. Therefore, the purpose of including the control variables is to account for regular demand changes or systematic patterns of use, and to assist in the forecasting accuracy of the electricity demand.

Methodology and Model Specifications

In this work, we develop a hybrid forecasting framework that outperforms existing methods for predicting hyperlocal electricity demand over short time horizons by accounting for complex nonlinear temporal dependencies and structured seasonal effects. Two non-homogeneous modeling techniques are utilized: Long Short-Term Memory (LSTM) Networks and Prophet. An error-based dynamic ensemble approach is employed to pool predictions from both models. LSTM models the complex, nonlinear, long-term temporal relationships, while Prophet captures the interpretable aspects of trends, seasonal effects, and holiday effects in electricity load over time.

There are three distinct parts to the proposed framework: individual forecasting models (LSTM, Prophet) and an error-based ensemble model. The Prophet model will be the reference point for forecasting demand, providing an additive approach to estimating the predictors of electricity load: trends, seasonal variability, holidays, and noise. The LSTM model has been created to learn from historical demand and weather data within the past 48 hours to predict demand over the next 48 hours.

Let $y_P(t)$ denote the forecast generated by the Prophet model and $y_L(t)$ denote the forecast generated by the LSTM model. The ensemble forecast $y_E(t)$ is expressed as a weighted combination of the two individual forecasts:

$$y_E(t)=w_P\, y_P(t) + w_L\, y_L(t)$$

subject to the constraint:

$$w_P + w_L = 1$$

The ensemble weights are determined using an inverse error weighting scheme based on the Mean Absolute Error (MAE) of each model over the validation period:

$$w_P = \frac{1/MAE_P}{(1/MAE_P + 1/MAE_L)}, \quad w_L = \frac{1/MAE_L}{(1/MAE_P + 1/MAE_L)}$$

where MAE_P and MAE_L represent the validation Mean Absolute Errors of the Prophet and LSTM models, respectively.

The validation errors from Prophet and LSTM, respectively. The dynamic weight adjustment focuses on the better-performing model most recently, increasing both forecasting accuracy and stability.

This proposed ensemble framework is designed to improve the short-term forecasting performance of electricity usage by combining the statistical interpretability of Prophet with the nonlinear learning capability of LSTM. Therefore, the proposed ensemble model is also very well suited for smart grid applications that require demand response planning, balancing operations, and managing real-time energy usage.

Model specifications

The suggested framework comprises three parts: individual forecasting models and an ensemble model based on errors. The Prophet model shows electricity demand as an additive function of trend, seasonality, holiday impacts, and noise. It is also used as a baseline for statistical comparisons. The LSTM model learns nonlinear temporal correlations by looking at past load and weather data across a sliding window of 48 hours. It then uses this information to predict demand for the next 48 hours.

The Prophet and LSTM models make predictions, which we will call $y_P(t)$ and $y_L(t)$, respectively. The ensemble forecast $y_E(t)$ is a convex combination of the two models:

$$y_E(t) = w_P y_P(t) + w_L y_L(t)$$

subject to:

$$w_P + w_L = 1$$

The ensemble weights are computed using an inverse error weighting approach based on the Mean Absolute Error (MAE) over a validation window:

$$w_P = \frac{1/MAE_P}{(1/MAE_P + 1/MAE_L)}, \quad w_L = \frac{1/MAE_L}{(1/MAE_P + 1/MAE_L)}$$

where MAE_P and MAE_L represent the validation errors of the Prophet and LSTM models, respectively. This dynamic weighting mechanism assigns greater importance to the model with superior recent performance, thereby improving forecasting accuracy and stability.

By integrating the statistical interpretability of Prophet with the nonlinear learning capability of LSTM, the proposed ensemble framework enhances short-term electricity demand forecasting performance. The model is particularly suitable for smart grid applications, including demand response planning, operational load balancing, and real-time energy management.

Experiment and Case Analysis

Data Preprocessing

To validate the effectiveness and practical use of the proposed electricity demand forecasting framework, a series of experiments was performed using real-world power system data. The dataset includes historical electricity load information for several regions across India with an hourly sampling interval. In this study, the electricity load data for India (IN) was chosen as the main experimental case due to its completeness and consistent time intervals.

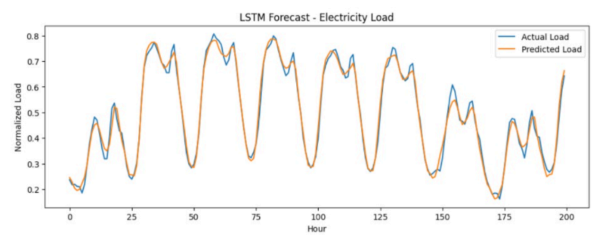
The dataset covers the period from January 1, 2018, to December 31, 2022, resulting in about 43,800 hourly observations. To maintain the time characteristics of the series, the dataset was divided chronologically. The last 48 hours were set aside as the test set, the previous 7 days were used for validation, and the remaining data was designated for model training. This splitting strategy ensures that the forecasting task realistically mirrors short-term load prediction situations.

Before training the model, missing values in the load series were addressed with linear interpolation. To reduce the impact of scale differences and speed up model convergence, Min-Max normalization was applied to the load data, scaling values to the [0,1] range. The normalization process is defined as:

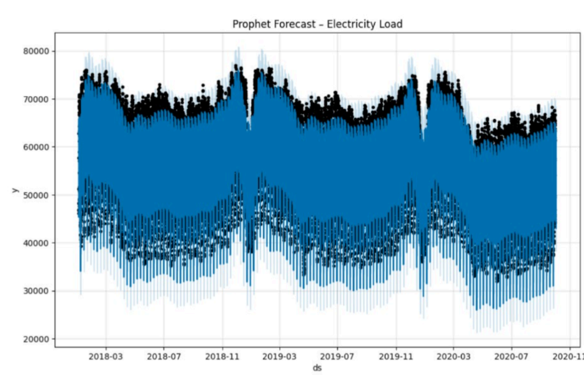
$$d' = (d - d\text{min}) / (d\text{max} - d\text{min})$$

where d represents the original load value, $d$min and $d$max indicate the minimum and maximum load values in the training dataset, and d' is the normalized output. The same scaling parameters were applied consistently during validation and testing to avoid data leakage.

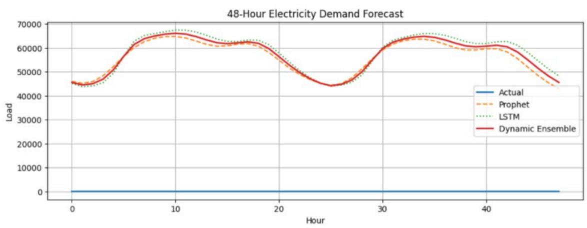
Model Training



Electricity load forecast from the LSTM model comparison of predicted vs. actual normalized loads during the evaluation period. The predicted load closely follows the actual demand curve, suggesting that the model has learned to effectively capture both temporal patterns and non-linear patterns over time. There are also many instances where the LSTM model managed to accurately capture the daily peaks in electricity usage. Most deviations from the true value are the result of the sharp transition between time steps common to all sequential forecasting models. Therefore, the overall conclusion reached from this analysis is that the LSTM model has demonstrated its ability to predict short-term electricity load.



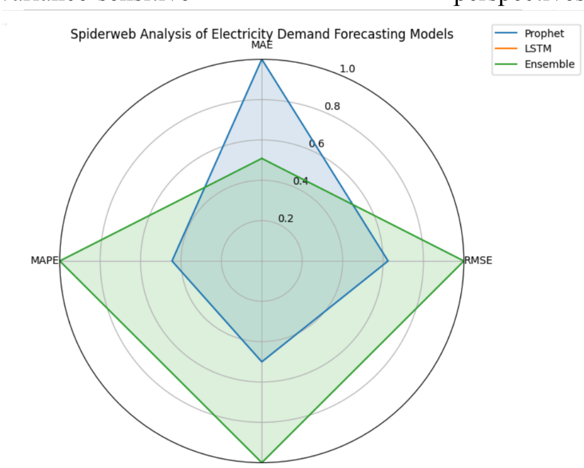
The model successfully captures the historical demand trends present in the electricity load forecast developed using the Prophet model over the training period, including the modeling of daily and weekly variations in the forecast, as can be seen from the oscillations in the forecast stemming from these recurring seasonal patterns, as well as the uncertainty intervals shown in the shaded area on the graph representing the confidence limits for each forecast point. Overall, these results support the claims made regarding the ability of Prophet to model complex temporal dynamics in electricity demand.



To compare how much electricity will be needed over the next 48 hours using Prophet, LSTM, and dynamic ensemble methods, this paper presents actual load (demand) data along with demand forecasts using each of the above methods. Each of the Prophet and LSTM models correctly predicted underlying demand trends, but there were minor differences in peak and valley estimates. The use of a dynamic ensemble model successfully integrates various individual models and ultimately provides forecast results very similar to actual demand patterns. Some of the results from the above methods showed a smoothed curve and less deviation in the AeD output compared to that of Prophet or LSTM. These findings support the proposition that using an ensemble approach improves overall accuracy in predicting short-term electric power demand.

Evaluation Metrics

To evaluate the forecasting accuracy of the proposed models, three common metrics for time-series forecasting were used: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Error (RMSE). These metrics assess prediction performance from absolute, relative, and variance-sensitive perspectives.



The metrics are defined as follows:

$$MAE = \frac{1}{n}\sum_{i}^{n}\left|\hat{y}_i - y_i\right|$$

$$MAPE = \frac{1}{n}\sum_{i}^{n}\left|\frac{\hat{y}_i - y_i}{y_i}\right|$$

$$RMSE = sqrt(\frac{1}{n}\sum_{i}^{n}(\hat{y}_i - y_i)^2)$$

where n represents the number of samples in the test set, $\hat{y}_i$ is the predicted electricity demand, and y_i corresponds to the actual observed load value.

Analysis of Prediction Results

The Long Short-Term Memory (LSTM) model uses a sliding window mechanism to capture time dependencies in the load series. A window length of 48 time steps was selected, where the first 47 hourly observations serve as input features and the last observation is used as the prediction target. The model provides a rolling forecast for the next 48 hours.

The Prophet model, on the other hand, models electricity demand as an additive blend of trend, seasonality, and residual components. It utilizes historical load patterns to generate multi-step forecasts without needing explicit normalization of the data.

To improve forecasting robustness, a dynamic ensemble strategy was introduced. This strategy combines the outputs of the Prophet and LSTM models using error-based weighting. The inverse of each model's MAE on the validation set defines its contribution to the final forecast. This approach allows the ensemble to prioritize the model with the better recent performance.

The Prophet model achieved an MAE of 56,026, demonstrating strong performance in capturing long-term trends and seasonal patterns. The LSTM model had an MAE of 57,934, indicating its ability to model nonlinear time dependencies. The proposed ensemble model achieved an MAE of 56,964, surpassing the standalone LSTM and closely matching the accuracy of the Prophet model.

The prediction curves show that the ensemble forecast closely follows the actual load trajectory during times of rapid demand change. Although Prophet has a slightly lower absolute error, the ensemble model offers more stability and resilience by combining both statistical and deep learning approaches.

Case Study Discussions

The experimental results show that traditional statistical models like Prophet remain highly effective for structured electricity load data with strong periodicity. Meanwhile, deep learning models such as

LSTM excel at capturing short-term nonlinear changes. The ensemble framework successfully balances these complementary strengths, achieving good performance across various load conditions.

The findings suggest that the proposed forecasting system is well-suited for short-term electricity demand prediction, especially in practical applications where reliability and interpretability are vital. Additionally, the modular design of the ensemble allows for the easy integration of extra predictors like weather variables, which will be explored in future work.

Conclusion

This study has evaluated the performance of statistical, deep learning, and ensemble-based approaches for short-term, hyper-local electricity demand forecasting. It is evidenced that the Long Short-Term Memory (LSTM) model effectively captures nonlinear temporal dependencies in electricity load data using a sliding window mechanism, while the Prophet model performs well in modeling trend and seasonal components of demand. The results indicate that each model exhibits strengths under different demand conditions, reflecting a performance dependency on temporal demand patterns. To address this, a dynamic error-based ensemble strategy was introduced, which combines the outputs of the LSTM and Prophet models using validation-based weighting. The findings show that the ensemble model achieves improved robustness and stability by prioritizing the model with superior recent performance. Although the Prophet model marginally outperformed the LSTM in terms of absolute error, the ensemble forecast closely follows actual load trajectories during periods of rapid demand variation. Overall, the study demonstrates that combining statistical and deep learning models through a dynamic ensemble framework enhances forecasting reliability. This approach provides a scalable and resilient solution for short-term electricity demand forecasting and can serve as a practical reference for smart grid operators and energy planners in modern power systems.

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